

ITSOLERA Pvt. Ltd.

Machine Learning Internship Project

Project Title: Traffic Volume Prediction System Using Machine Learning

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Abstract:

This project presents a Machine Learning-based Traffic Volume Prediction System developed to forecast hourly traffic volume using historical traffic and environmental data. Multiple regression models including Linear Regression, Decision Tree, Random Forest, Support Vector Regression (SVR), and XGBoost were implemented and evaluated. The system applies feature engineering and preprocessing techniques to improve predictive performance. Ensemble-based models demonstrated superior accuracy compared to traditional regression models.

1. Introduction:

Urban traffic congestion significantly affects transportation efficiency and economic productivity. Accurate traffic prediction enables transportation authorities to optimize traffic management, reduce congestion, and improve road planning strategies.

2. Problem Statement:

To develop a regression-based machine learning model capable of accurately predicting traffic volume using time-based and environmental features.

3. Dataset Description:

The project uses the Metro Interstate Traffic Volume dataset. The dataset includes features such as holiday, temperature, weather conditions, and traffic volume (target variable).

4. Methodology:

4.1 Data Preprocessing

- Handled missing values in the holiday column.
- Dropped unnecessary columns such as weather_description.
- Encoded categorical variables using LabelEncoder.
- Split dataset into training and testing sets (80/20).

4.2 Feature Engineering

- Created weekend indicator feature.
- Encoded weather and holiday features.

- Selected input (X) and output (y) variables.

4.3 Model Development

The following regression models were implemented:

- Linear Regression
- Decision Tree Regressor
- Random Forest Regressor
- Support Vector Regressor (SVR)
- XGBoost Regressor

4.4 Model Evaluation

Models were evaluated using the following metrics:

- R^2 Score
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- Mean Absolute Error (MAE)

An Actual vs Predicted scatter plot was generated to visualize performance.

5. Results:

Among all implemented models, ensemble techniques such as Random Forest and XGBoost provided better performance compared to Linear Regression. Decision Tree showed signs of overfitting on training data. The final selected model was Random Forest, which was saved using Pickle for future deployment.

First 5 rows:

	holiday	temp	rain_1h	snow_1h	clouds_all	weather_main	weather_description	date_time	traffic_volume
0	NaN	288.28	0.0	0.0	40	Clouds	scattered clouds	2012-10-02 09:00:00	5545
1	NaN	289.36	0.0	0.0	75	Clouds	broken clouds	2012-10-02 10:00:00	4516
2	NaN	289.58	0.0	0.0	90	Clouds	overcast clouds	2012-10-02 11:00:00	4767
3	NaN	290.13	0.0	0.0	90	Clouds	overcast clouds	2012-10-02 12:00:00	5026
4	NaN	291.14	0.0	0.0	75	Clouds	broken clouds	2012-10-02 13:00:00	4918

```

Dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 48204 entries, 0 to 48203
Data columns (total 9 columns):
#   Column                Non-Null Count  Dtype
---  -
0   holiday                61 non-null    object
1   temp                  48204 non-null float64
2   rain_1h               48204 non-null float64
3   snow_1h               48204 non-null float64
4   clouds_all            48204 non-null int64
5   weather_main          48204 non-null object
6   weather_description   48204 non-null object
7   date_time             48204 non-null object
8   traffic_volume        48204 non-null int64
dtypes: float64(3), int64(2), object(4)
memory usage: 3.3+ MB
None

```

```

Missing Values per Column:
holiday                48143
temp                  0
rain_1h               0
snow_1h               0
...
weather_description   0
date_time             0
traffic_volume        0
dtype: int64

```

```

Index(['holiday', 'temp', 'rain_1h', 'snow_1h', 'clouds_all', 'weather_main',
      'traffic_volume', 'hour', 'day', 'month', 'day_of_week', 'is_weekend'],
      dtype='object')

```

Cleaned & Feature Engineered Dataset:

	holiday	temp	rain_1h	snow_1h	clouds_all	weather_main	traffic_volume	hour	day	month	day_of_week	is_weekend
0	7	288.28	0.0	0.0	40	1	5545	9	2	10	1	0
1	7	289.36	0.0	0.0	75	1	4516	10	2	10	1	0
2	7	289.58	0.0	0.0	90	1	4767	11	2	10	1	0
3	7	290.13	0.0	0.0	90	1	5026	12	2	10	1	0
4	7	291.14	0.0	0.0	75	1	4918	13	2	10	1	0

```
Train Shape: (38563, 11)
Test Shape: (9641, 11)
```

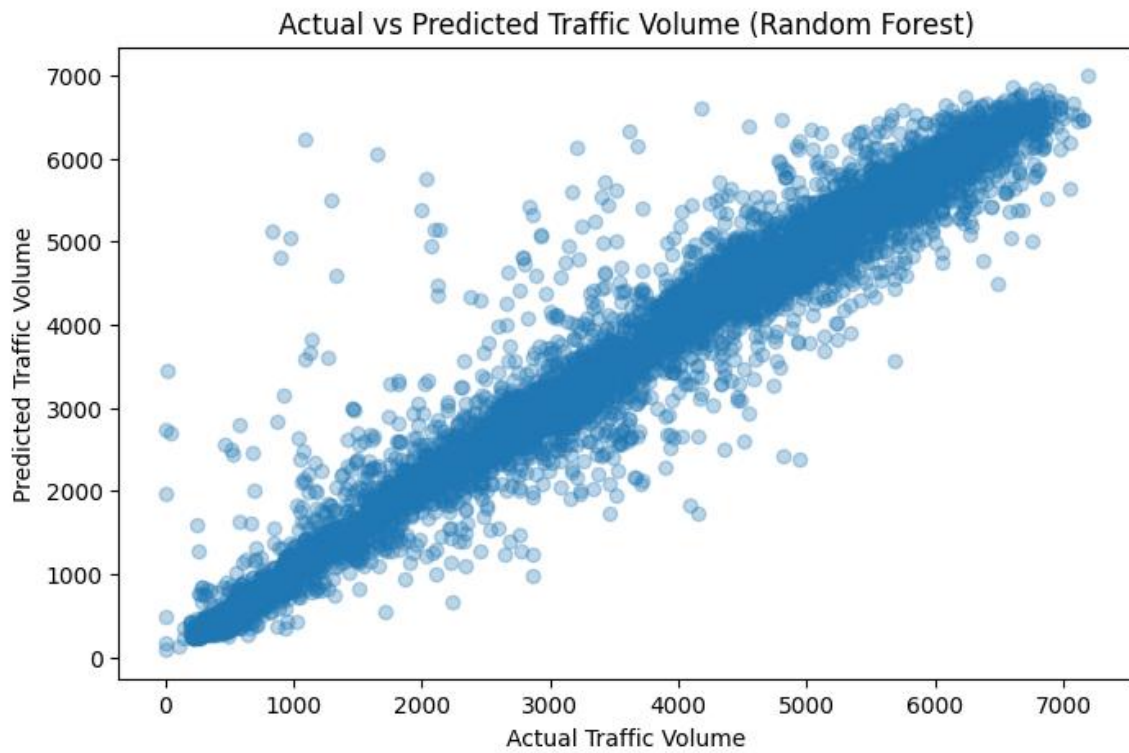
R2 Score on Training Data:

```
Linear Regression: 0.18742395493791864
Decision Tree: 1.0
Random Forest: 0.9950202599158389
SVR: 0.0066334022241235635
XGBoost: 0.9761314392089844
```

```
Test R2 Score (Random Forest): 0.9651593265461431
Mean Squared Error: 137743.074907354
Root Mean Squared Error: 371.1375417649823
Mean Absolute Error: 208.94576911108805
```

Comparative Analysis of Regression Models:

Model Name	Training R2 Score	Test R2 Score	MAE	RMSE	Performance Status
Linear Regression	0.187	0.185	High	High	Underfit
SVR	0.006	0.005	Very High	Very High	Poor
Decision Tree	1.000	0.941	Low	Low	Overfit
Random Forest	0.995	0.965	208.94	371.13	Best Model
XGBoost	0.976	0.958	Low	Low	Good



6. Tools and Technologies:

Programming Language: Python

Libraries: Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, XGBoost

Development Environment: Vscode, Python and Jupyter Notebook

Model Saving: Pickle

7. Challenges I Faced:

- Handling missing values
- Encoding categorical features
- Avoiding overfitting in Decision Tree
- Performance tuning of models

8. Conclusion:

The Traffic Volume Prediction System successfully demonstrates the practical application of machine learning regression models in urban traffic analytics. Ensemble learning methods improved prediction accuracy. The project provided valuable hands-on experience in data preprocessing, feature engineering, model training, evaluation, and deployment preparation.